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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CCT111 – SPECIAL CONCRETES

(For CCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: i) Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

ii) Use of IS456-2000 and IS-10262-2019 is permitted.

iii) Assume missing data suitably and mention the same.

Unit – I

	Marks	BT*	CO*	PO*
1. Develop the mix proportions of Structural Light weight concrete for target strength 24MPa.	20	L3	2	1,2
2. a) Explain how the development of cement replacement materials such as fly ash, silica fume, and rice husk ash contributes to sustainable concrete production.	10	L2	1	1,2
b) What is high-density concrete? Explain its significance in radiation shielding and list the materials used.	10	L2	2	1,2
3. Develop the concrete mix proportions for a specified 28-day target compressive strength of 30 MPa using 20 mm maximum size coarse aggregate as per relevant IS code provisions.	20	L3	1	1,2

Unit – II

4. Develop the M 35 Grade SCC, Assuming all necessary data suitably.	20	L3	3	1,2
5. Develop the M80 Grade ECC. Assume necessary data suitably.	20	L3	4	1,2
6. a) Explain any three testing methods used to assess the fresh properties of Self-Compacting Concrete.	10	L2	3	1,2
b) Explain the Engineered Cementitious Composites (ECC). Explain the role of fibers and the crack-arrest mechanism in ECC.	10	L2	4	1,2

Unit – III

7. Develop the mix proportion for M-70 Grade concrete. Assume necessary data suitably.	20	L3	5	1,2
8. Apply the concept of self-healing concrete and evaluate the challenges faced during large-scale use of bacterial concrete. Construct a set of practical solutions to overcome these challenges.	20	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
